

PROJECT STORMSIGHT

WALMART'S SUPPLY CHAIN OPTIMIZATION DURING EXTREME WEATHER

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EXECUTIVE SUMMARY

CORE PROBLEM

Walmart's current supply chain reacts to storm-driven demand after it occurs, leading to:

- stockouts at high-risk stores
- overloading at specific distribution centers
- missed revenue during peak demand periods

ANALYSIS OBJECTIVES

- Identify when demand spikes occur relative to storm proximity
- Determine which products drive the majority of excess demand
- Identify which distribution center lanes are most at risk of stockouts
- Recommend how Walmart should shift supply chain resources before demand peaks

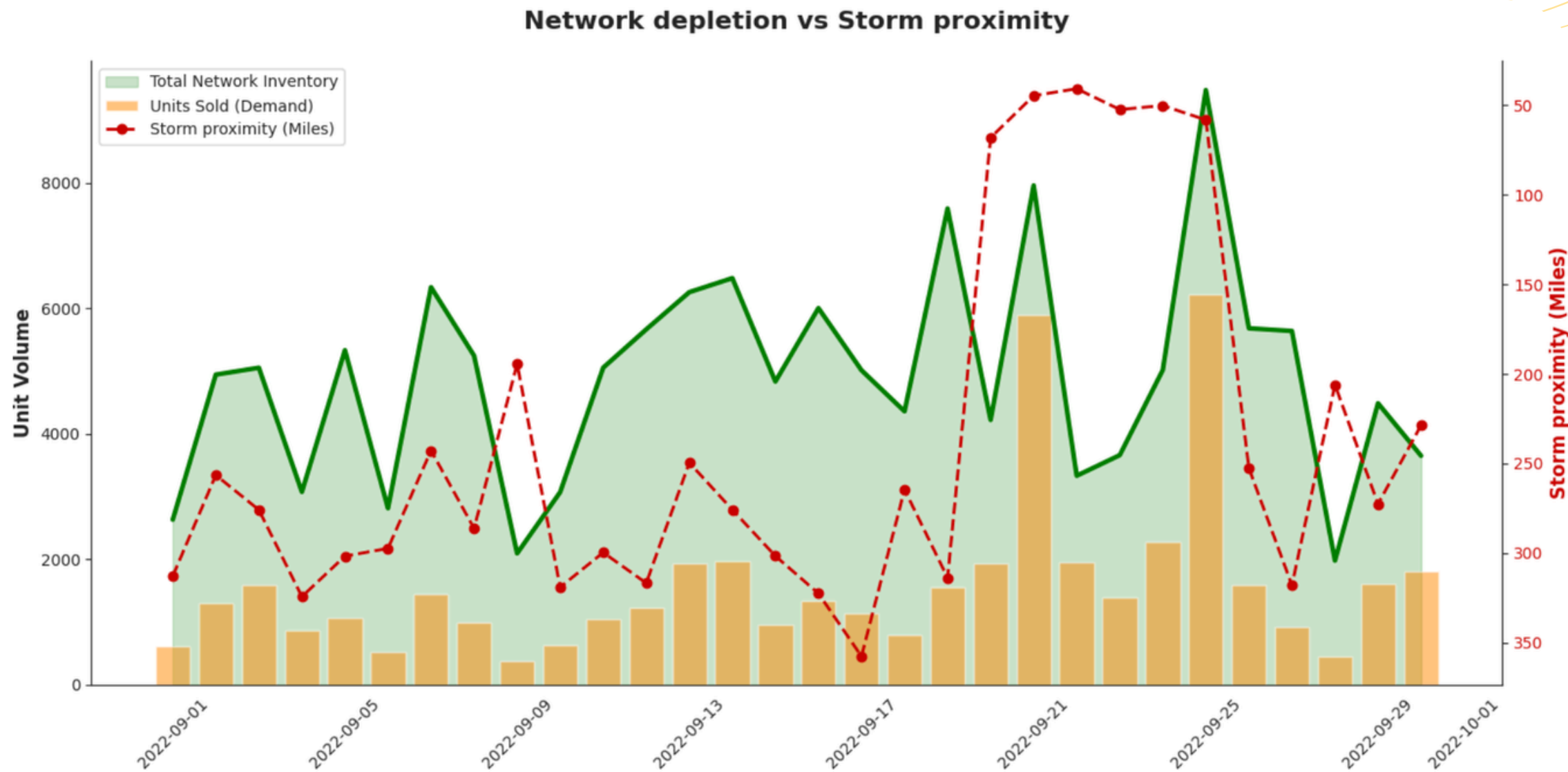
KEY TAKEAWAYS

Walmart should implement a priority-based rerouting system that:

- shifts supply to high-risk stores before inventory depletion
- prioritizes a small set of high-impact products
- redistributes load away from overloaded distribution centers
- accelerates fulfillment of essential items during demand surges



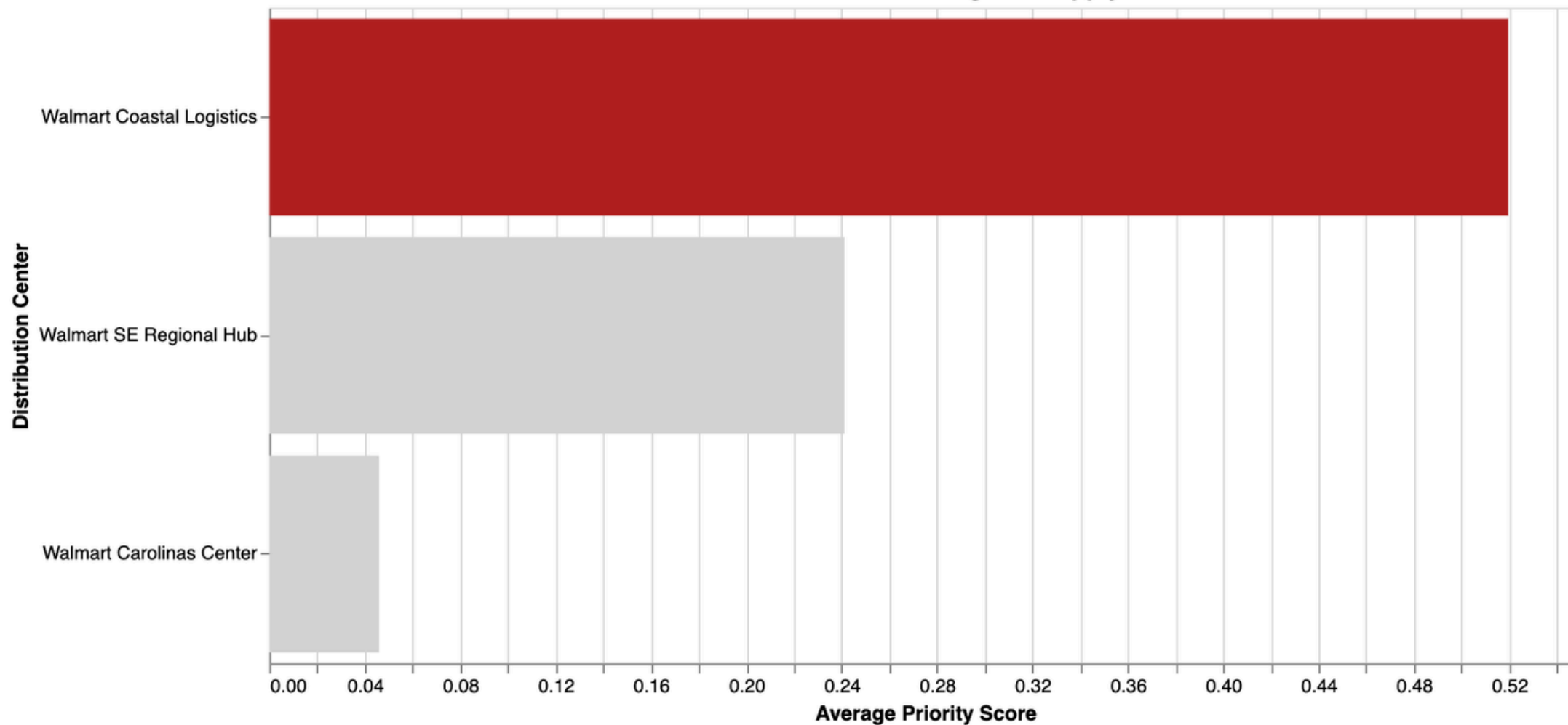
KEY FINDING #1



- Minor demand spikes on Sept 7 and Sept 13. When the storm was 250+ miles away, consumers began preemptive buying.
 - As the storm proximity was below 100 miles, "Units sold" (orange bars) spiked to their highest levels in the 30 days.
 - Total Network Inventory (green area) was aggressively ramped up between Sept 23 and Sept 25, peaking at nearly 10,000 units.
- Indicates a transition from a "pre-storm hoarding" phase to an "impact/recovery" phase, where store accessibility or consumer movement may have been restricted.

KEY FINDING #2

One Distribution Center Faces the Highest Supply Chain Pressure



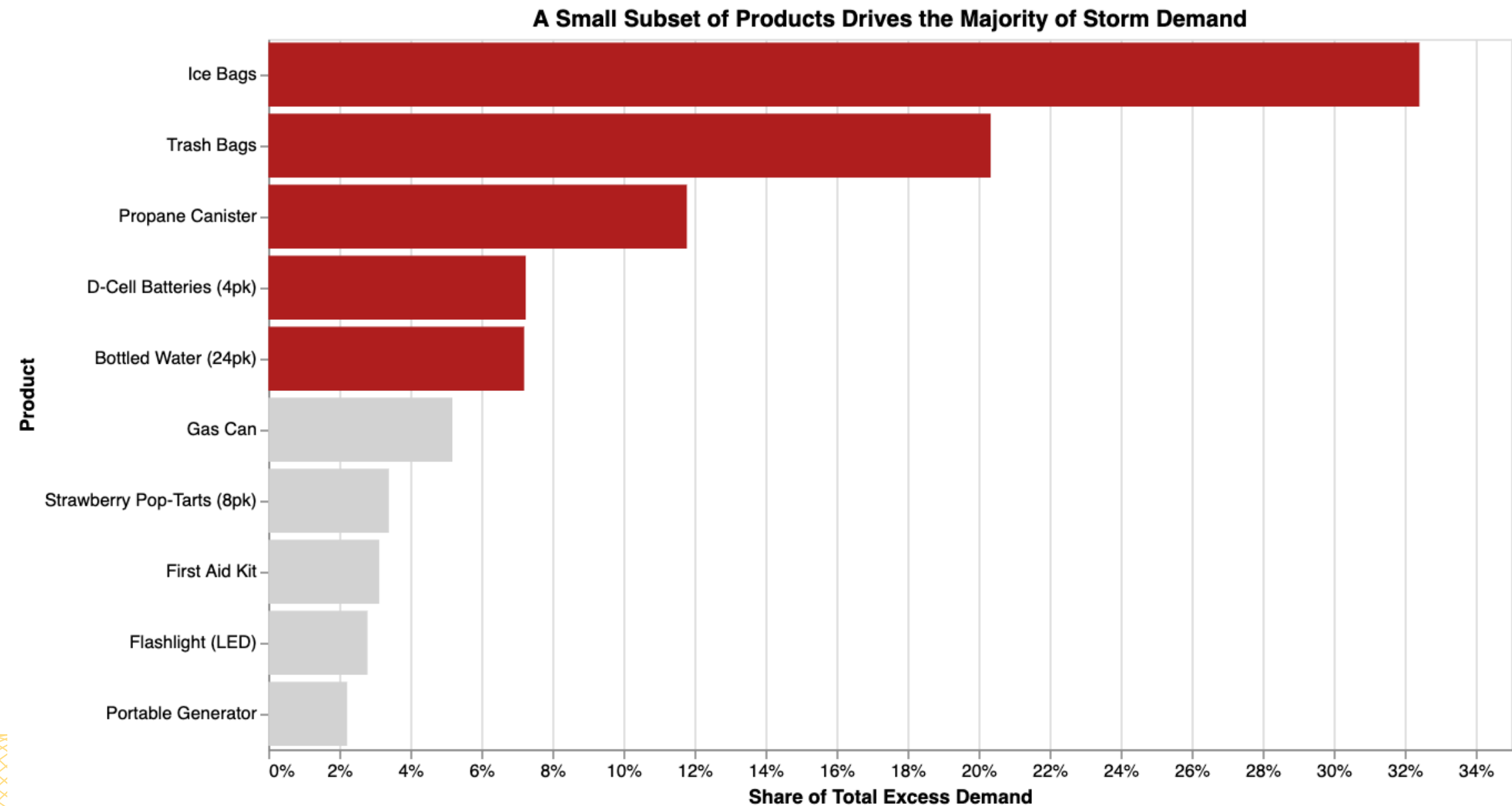
- Walmart Coastal Logistics burden is more than double that of the SE Regional Hub and over ten times that of the Carolinas Center.
- The majority of storm-related demand is being channeled through a single facility, creating a potential "choke point" region.

→ Represents a significant opportunity for load-balancing or rerouting supply chain lanes to reduce pressure from the Coastal center

KEY FINDING #3

- Top 5 products = 79% of total excess demand
- Top 10 = over 95% of total excess demand
- High-demand items such as ice bags, trash bags, and propane canisters dominate excess demand
→ consumers prioritize immediate preparation and survival needs as storms approach

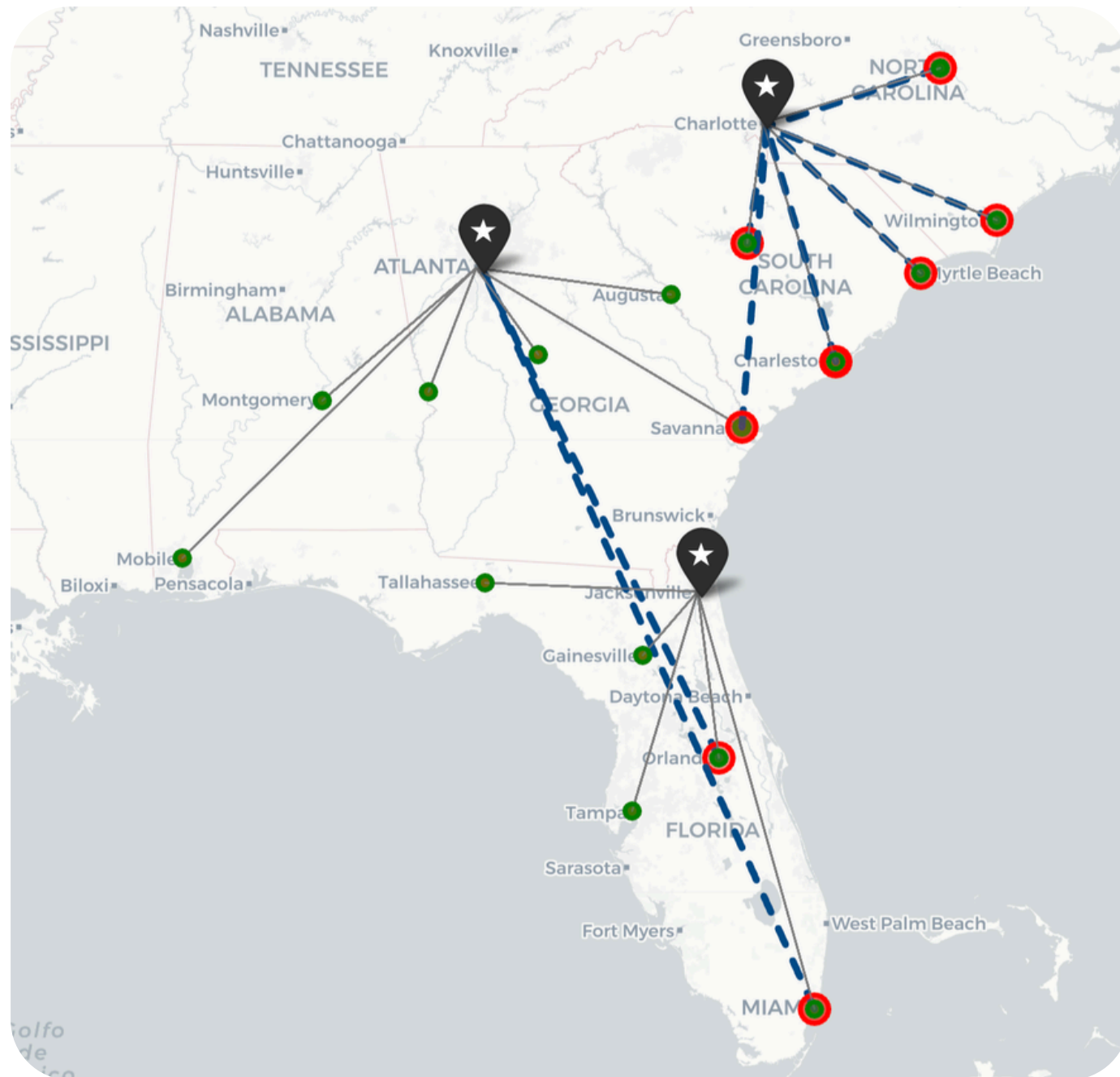
→ Presents significant opportunity to streamline supply chain operations by focusing inventory allocation, transportation capacity, and pre-staging on a small set of products rather than distributing resources broadly



REROUTING MODEL

- The stars & Circles: The Black Stars are our Regional Hubs (Distribution Centers). The Circles are our stores.
- The Intensity (Red Dots): A Red Circle indicates a store where the Priority Score is in the top 15%—meaning they have high demand, low inventory, and a storm is closing in.
- The Lanes (Solid vs. Dashed): * Gray Lines: Standard supply chain routes (Business as Usual).
- Blue Dashed Lines: The "Emergency Flex." These represent the proposed shift in fulfillment from coastal hubs to inland safe-havens (Atlanta/Charlotte).

→ The Rerouting Model demonstrates that shifting specific coastal lanes to inland hubs prevents the Walmart Coastal Logistics center from being overloaded(can help implementing a supply chain management system that triggers when a store Priority Score > 85th Percentile AND Location East of -81.5 Longitude.)



SOLUTIONS

RECOMMENDATIONS

TRIGGER-BASED REROUTING SYSTEM

Action

- Activate rerouting when priority score exceeds a threshold of 15% (identifies high risk store-product lanes)
- Shift supply from Coastal DC to inland hubs (supply chain pressure is 2x higher at Coastal DC)

Why

- Demand spikes are predictable 72–96 hours in advance, allowing for intervention before stockouts occur
- Leverages Walmart's national DC network and routing flexibility
- Addresses current capacity imbalance

RECCOMENDATIONS

PRE-STAGE HIGH IMPACT PRODUCTS

Action

- Pre-position top 5 products at regional hubs (demand for top 5 products accounts for around 79% of total excess demand)
- Prioritize these SKUs in transportation and inventory allocation (demand is highly concentrated, not evenly distributed)

Why

- Leverages strong supplier relationships and inventory scale
- Capitalizes on the opportunity to capture crisis driven-demand and increase loyalty
- Broad inventory strategies waste available capacity → targeted allocation allows for immediate gain

RECOMMENDATIONS

FAST-LANE FULFILLMENT FOR HIGH DEMAND ITEMS

Action

- Create “fast lanes” for high-impact SKUs (bypass standard DC sorting)
- Use pre-packed storm kits for faster store replenishment (demand spikes rapidly as storms get closer)
- Prioritize direct shipment from regional hubs to stores (high impact products need faster throughput to avoid stockouts)

Why

- Leverages existing high-volume logistics network and cross-docking capabilities
- Standard fulfillment processes are too slow during demand surges
- Faster execution allows Walmart to capture demand instead of losing it to stockouts

PRIMARY RISKS

DATA LATENCY

› Delays in real-time POS data can reduce Priority Score accuracy, leading to late or incorrect rerouting decisions

- Heavy reliance on real-time POS data + rapidly shifting demand patterns during storms

Mitigation

- Leverage real-time data infrastructure to increase refresh frequency as much as possible
- Implement threshold buffers and fallback rules so that sensitivity to data delays is reduced as much as possible

EXECUTION AND COORDINATION RISK

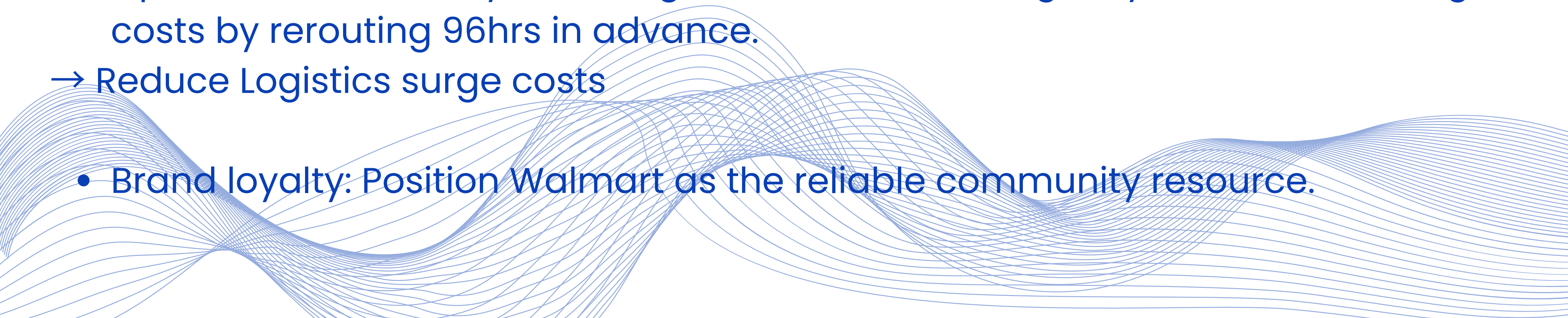
› Distribution centers and store operations may not respond quickly enough to rerouting recommendations during high demand windows

- Requires cross-DC coordination and a lot of trained individuals + time-sensitive conditions increase operational pressure

Mitigation

- Use Walmart's standardized logistics network and scale as much as possible
- Develop predefined rerouting protocols and training programs

POTENTIAL UPSIDE

- Revenue protection: capturing the 4x surge in emergency SKU demand rather than losing it to out-of-stocks.
→ Increase regional revenue
 - Operational efficiency: reducing overtime and emergency last-mile trucking costs by rerouting 96hrs in advance.
→ Reduce Logistics surge costs
 - Brand loyalty: Position Walmart as the reliable community resource.
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CLOSING REMARKS

By acting earlier and allocating smarter, Walmart can turn supply chain disruptions into a competitive advantage.

Optimizing supply chains during storms requires moving from:

- reacting to store-level stockouts → to anticipating demand and acting before it peaks
- distributing inventory broadly → to focusing resources where they matter most

What it allows:

- stay in stock when competitors run out
- strengthen long-term customer trust during critical moments

The background features a series of thin, light blue lines that flow and curve across the frame, creating a sense of movement and depth. These lines are most concentrated on the left side and bottom, where they form dense, overlapping waves that gradually thin out towards the right.

THANK YOU

APPENDIX A.

SWOT ANALYSIS FOR IMPLEMENTING REROUTING MODEL

Strengths (Internal Assets)

- The "Priority Score" provides a 72–96 hour head start over reactive competitors.
- Walmart's existing distribution network allows stores to be reassigned to secondary hubs without redesigning the logistics system.
- Real-time POS and inventory tracking systems enable continuous monitoring of demand spikes and stockout risk, supporting dynamic decision-making.

Weaknesses (Internal Constraints)

- The model is heavily dependent on real-time Point of Sale data. Any lag in store-level reporting can degrade the Priority Score accuracy.
- Coordinating rerouting decisions across DCs requires training and standardized protocols
- Shifting demand away from the Coastal DC creates a temporary strain on inland hubs (e.g., Atlanta, Charlotte), requiring short-term labor and scaling

Opportunities

- The core logic can be expanded to manage logistics during winter storms, wildfires,...
- The prescriptive output can be fed directly into future autonomous trucking routes for 24/7 replenishment.
- Being the only retailer with full shelves during a crisis, Walmart can capture "Crisis Shoppers" and convert them into long-term loyalists.

Threats

- Competitors investing in predictive logistics and rapid fulfillment can reduce Walmart's advantage
- Infrastructure disruptions can limit the execution of rerouting decisions, even with model accuracy

*Findings from Walmart 10-K filings, Statista (retail scale, logistics metrics),MarketLine / company research summaries

APPENDIX B.

KEY METRICS DRIVING SUPPLY CHAIN DECISIONS

PRIORITY SCORE

Definition: Metric measuring how urgently a store–product combination requires supply chain intervention

Calculation

- Combines three key drivers of risk:
 - Demand Spike Ratio → how quickly demand is increasing
 - Inventory Coverage Gap → how close the store is to running out of stock
 - Lead Time Risk → whether replenishment can arrive before stockout
 - *These components are normalized and combined to create a single score that reflects overall supply chain pressure.*

Importance: enables Walmart to prioritize limited supply chain capacity and drive targeted rerouting decisions instead of reactive replenishment

DEMAND SPIKE RATIO

Definition: Metric measuring how much demand increases relative to normal levels

Calculation

- Demand Spike Ratio = Units Sold ÷ Average Daily Sales

Importance: Identifies which products drive storm demand, enabling targeted inventory allocation

APPENDIX C.

Q: How do we know the Priority Score actually identifies real risk?

A:

- Combines demand spike, inventory coverage, and lead time risk — the three key drivers of stockouts
- High-priority cases align with observed low coverage (<48 hours) and high demand spikes
- This ensures the model reflects real operational risk, not just statistical variation

Q: Why use the top 15% threshold for rerouting decisions?

A:

- Priority Scores are highly skewed, with a small subset driving most risk
- The top 15% captures the upper tail where risk is concentrated
- This balances impact (focus on critical cases) and cost (avoid unnecessary rerouting)

APPENDIX D.

Q: Why does this strategy give Walmart an edge?

A:

- Most competitors rely on reactive replenishment
- Walmart can act 72–96 hours earlier, maintaining availability during peak demand
- This leads to higher revenue capture and stronger customer loyalty

Q: How do we know that Demand Spike Ratio and Inventory Coverage are the correct predictors of stockout risk?

A:

- The variables directly capture the primary drivers of stockouts: demand acceleration and available supply
- High spike ratios combined with low inventory coverage consistently align with imminent stockout scenarios

APPENDIX E.

Q: How robust is the model to rapid changes in storm trajectory or demand signals?

A:

- Model is designed to update dynamically whenever new data becomes available
- Threshold-based triggers and prioritizing ensure that the consistently high-risk cases drive decisions.
 - reduces sensitivity to short-term volatility